HR Integration

Expanded Descriptive Analysis Report

Workforce comparison of Unit 1 and Unit 2

18 visualizations • one graph per page • descriptive interpretations

Reference date: 31 December 2016

Source workbook: Data_Set_Case_Study_Homework(1).xlsx

This expanded report includes the original eight charts plus additional stacked bars, hiring and promotion charts, boxplots, a career-level horizontal bar chart, an education-mix chart, a time-in-job histogram, a violin plot and a normalized area chart. Every graph has its own dedicated page with explanatory text.

Decision caveat: This is a historical descriptive analysis. It does not prove role duplication, establish causes of exits, or quantify integration savings.

1. Executive summary

The supplied records show differences in workforce scale, age, tenure, career-level mix and termination patterns. Unit 1 has more active employees and higher median age and tenure. Unit 2 has 60 duplicate employee IDs requiring reconciliation, and its cleaned records include more terminated employees than Unit 1. Validate the period, definitions and duplicate handling before formal reporting.

Indicator	Unit 1	Unit 2
Raw employee rows	6,328	2,793
Unique IDs retained	6,328	2,733
Duplicate employee IDs	0	60
Active employees	5,920	2,180
Terminated records	408	553
Voluntary exits	323	172
Involuntary exits	85	381
Median age (active)	31.3 years	28.0 years
Median tenure (active)	8.7 years	5.0 years

Counts of terminated records are not annual turnover rates. A turnover rate requires a defined reporting period and an appropriate average-headcount denominator. The workbook does not include enough role, FTE and compensation detail to establish role overlap or cost savings.

2. Data preparation, assumptions and limitations

The workbook has separate Unit 1 and Unit 2 worksheets. The dashboard cleaning logic reads the employee table after introductory rows, parses dates and calculates age, tenure and time in job using 31 December 2016 as the reference date. Durations are calculated as elapsed days divided by 365.2425 and rounded to one decimal place.

Duplicate employee IDs are handled by retaining the row with the most populated date fields. Unit 2 has 60 repeated IDs (120 rows participate in duplicate IDs); Unit 1 has none under this rule. This is a practical rule, not proof that the retained row is authoritative. HR should reconcile the repeated IDs.

The workbook's status, termination type, hire-in-FY and promotion-in-FY flags are used as provided. Definitions and reporting periods should be confirmed before treating outputs as official HR metrics. Performance rating scales should not be compared across units unless their meaning and scales are harmonized.

The data does not provide sufficient job title, job family, department, location, reporting-line, FTE and compensation detail to confirm duplicated roles or estimate savings. The results describe the historical supplied data, not the current workforce.

Recommended next steps

• Reconcile duplicate employee IDs and confirm authoritative employee records.

• Validate the effective date, reporting period and definitions for employment status and exit types.

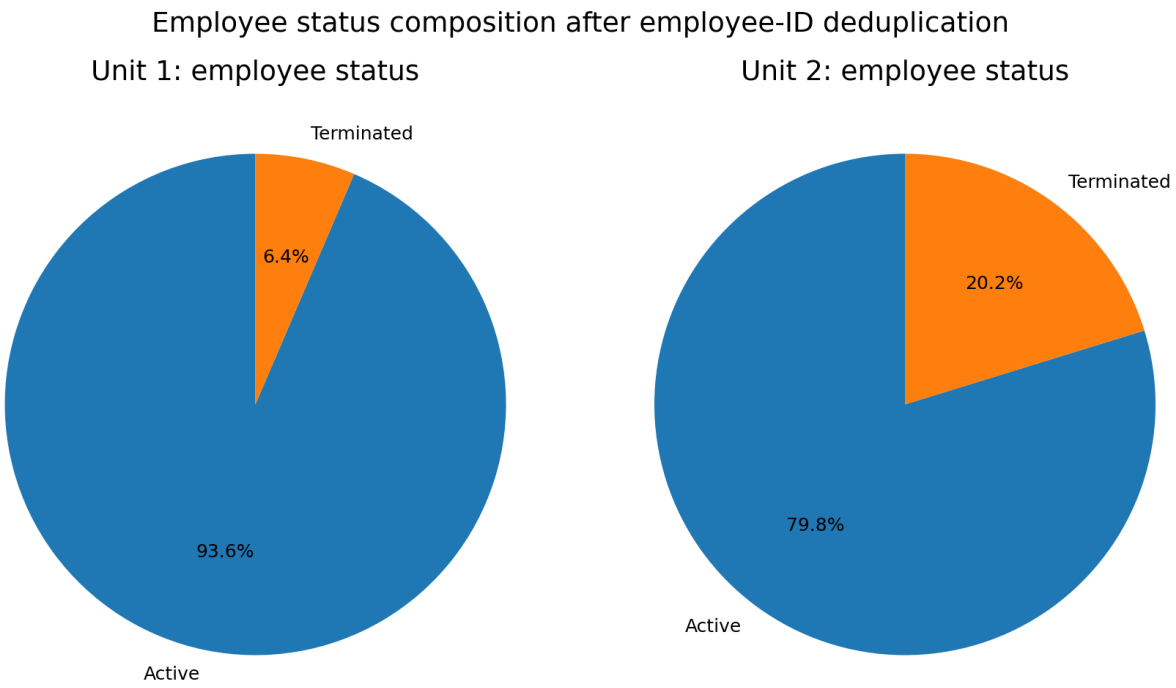
• Harmonize career-level and performance-rating definitions before comparing units.

• Obtain role, department, location, reporting-line, FTE and compensation data for integration modelling.

• Map critical roles and skills, retention risks, communications and change-management requirements.

3. Pie chart — employee status

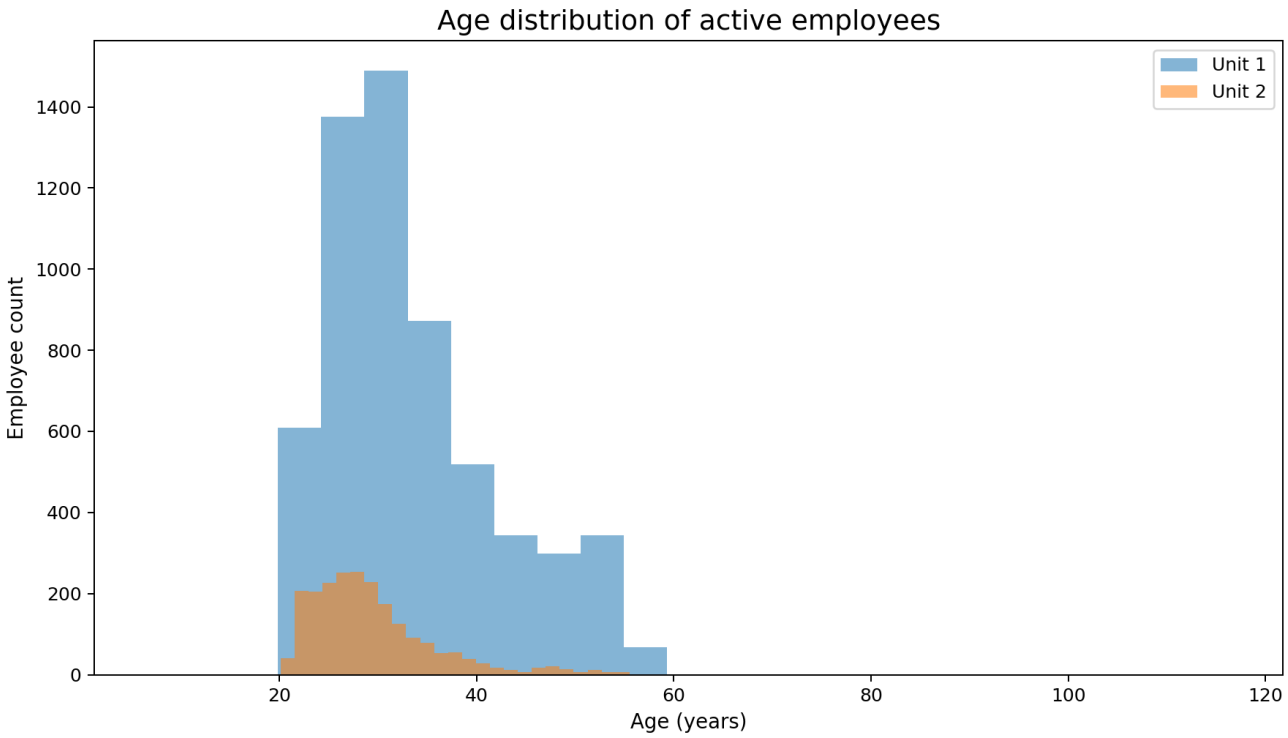
Compares the proportions of active and terminated records within each unit after employee-ID deduplication. These are historical record proportions, not annual turnover rates.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

4. Histogram — age distribution

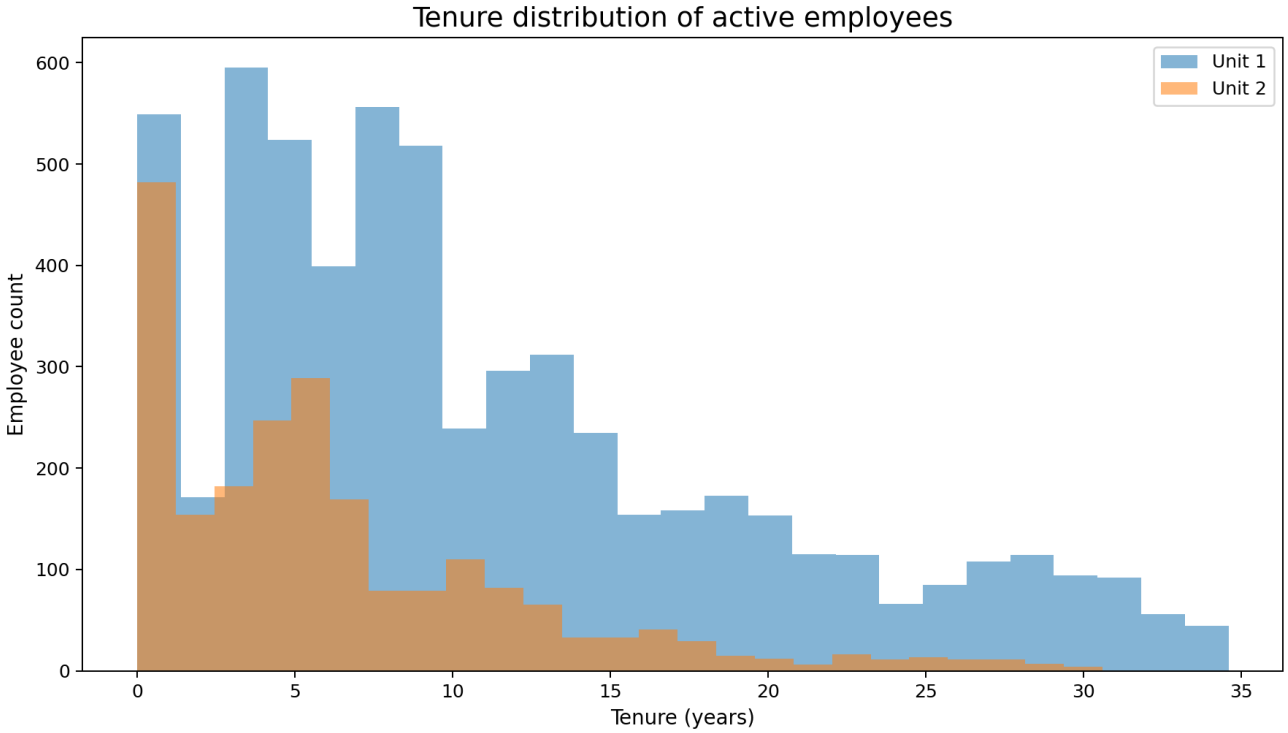
Compares calculated ages among active employees. Median age is 31.3 years in Unit 1 and 28.0 years in Unit 2. Age alone does not indicate performance or future retirement.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

5. Histogram — tenure distribution

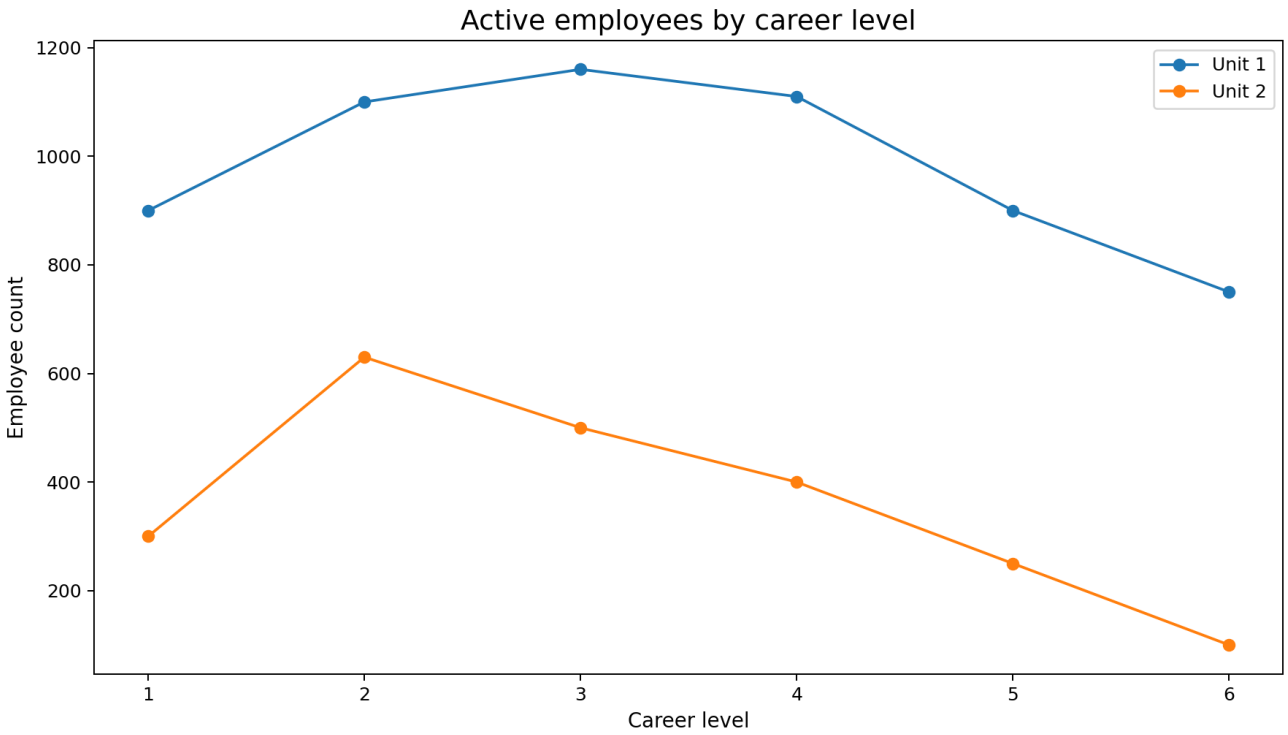
Compares calculated service years among active employees. Median tenure is 8.7 years in Unit 1 and 5.0 years in Unit 2.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

6. Line chart — active employees by career level

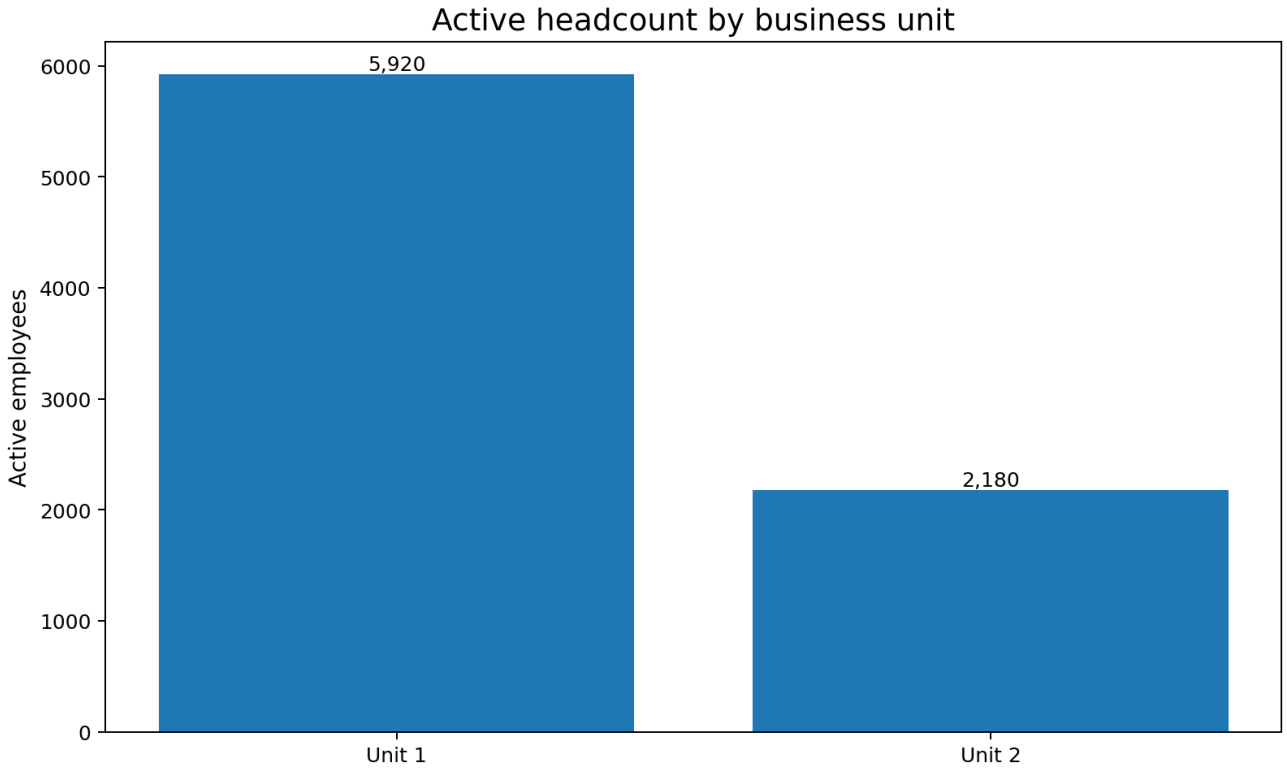
Connects employee counts across career levels to compare the two units. Career level is categorical, so this is not a time-series trend.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

7. Bar chart — active headcount

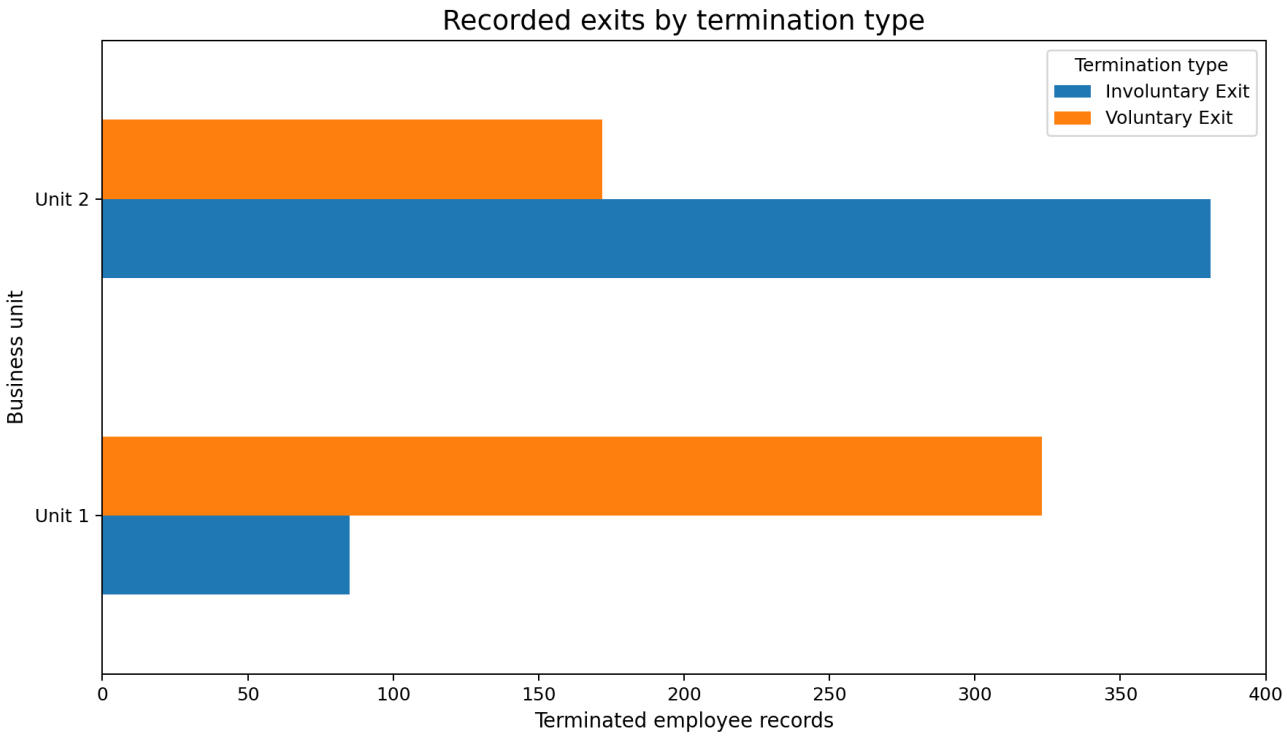
Shows active employee counts after duplicate-ID handling: 5,920 in Unit 1 and 2,180 in Unit 2. Headcount alone cannot establish overstaffing.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

8. Horizontal bar chart — exits by type

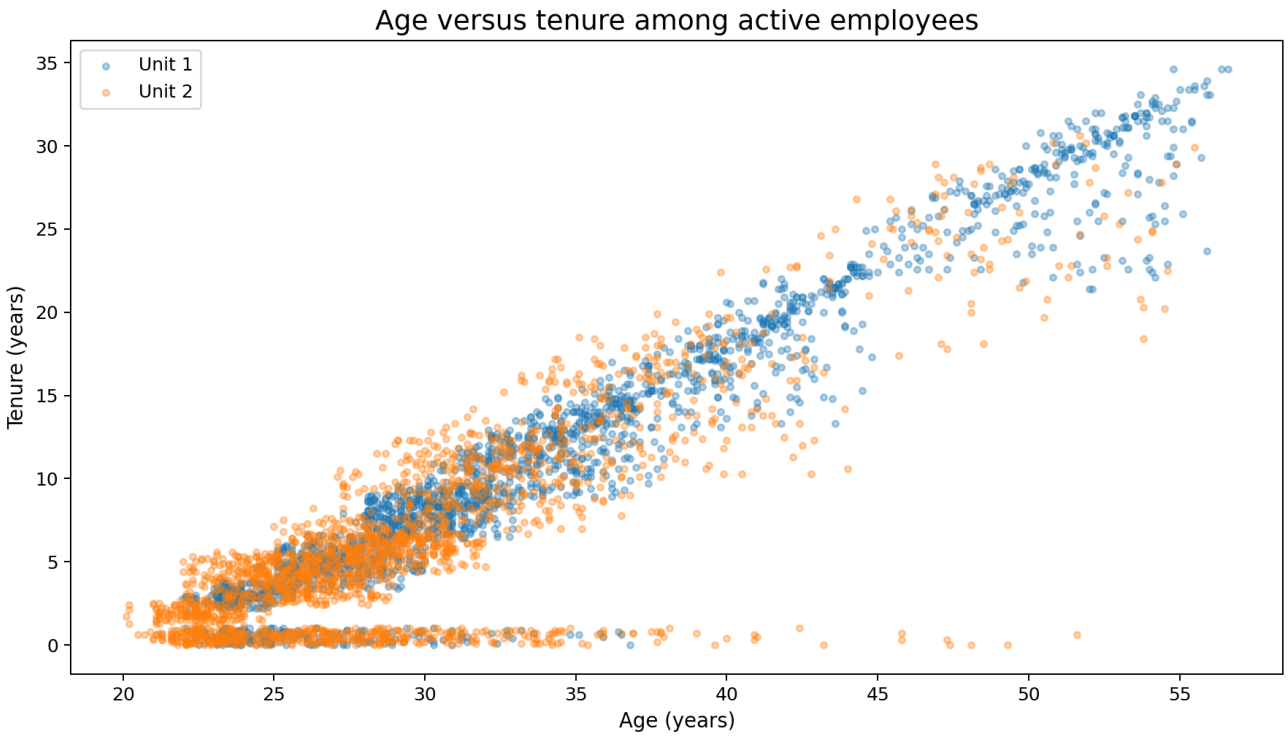
Compares recorded exits by termination type. Unit 1 has 323 voluntary and 85 involuntary exits; Unit 2 has 172 voluntary and 381 involuntary exits. Validate period coverage and definitions.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

9. Scatter plot — age versus tenure

Explores how age and tenure co-vary among active employees. Points are downsampled for readability. Association does not imply causation.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

10. Heatmap — correlations

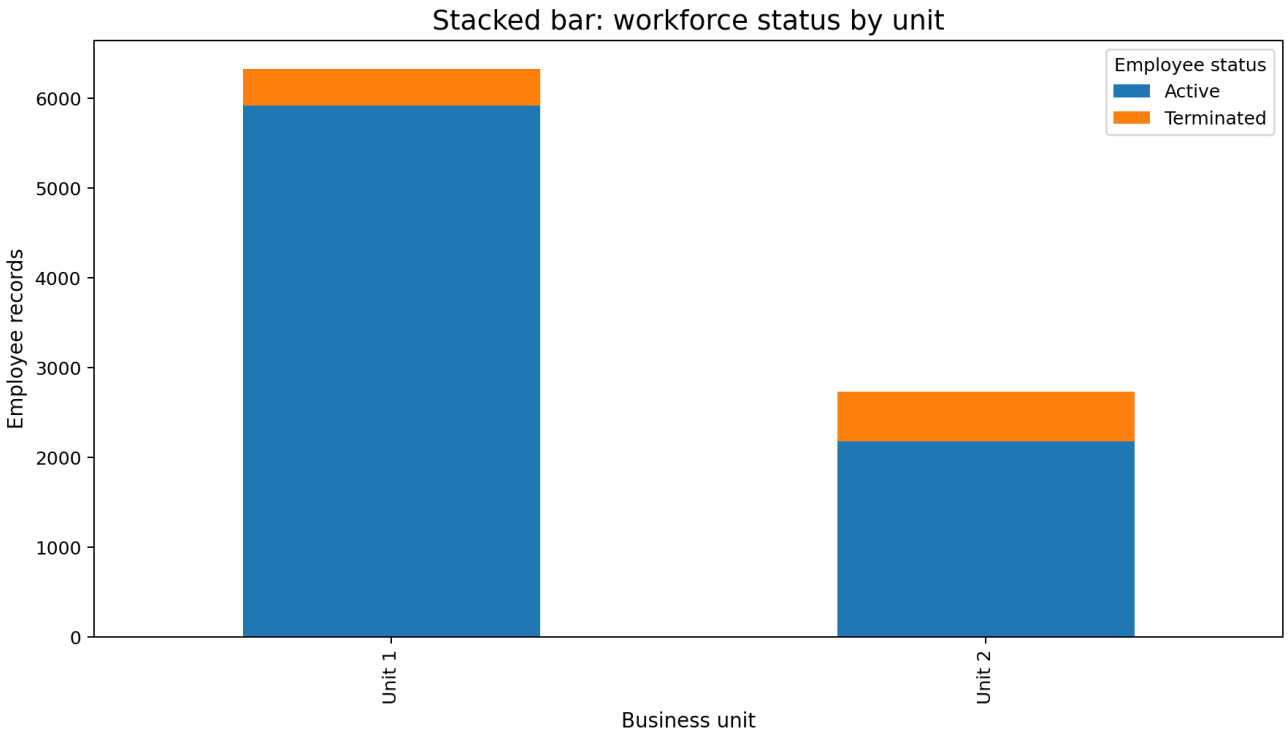
Shows Pearson correlations among career level, age, tenure and time in job. Correlations indicate association, not causation.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

12. Stacked bar — workforce status

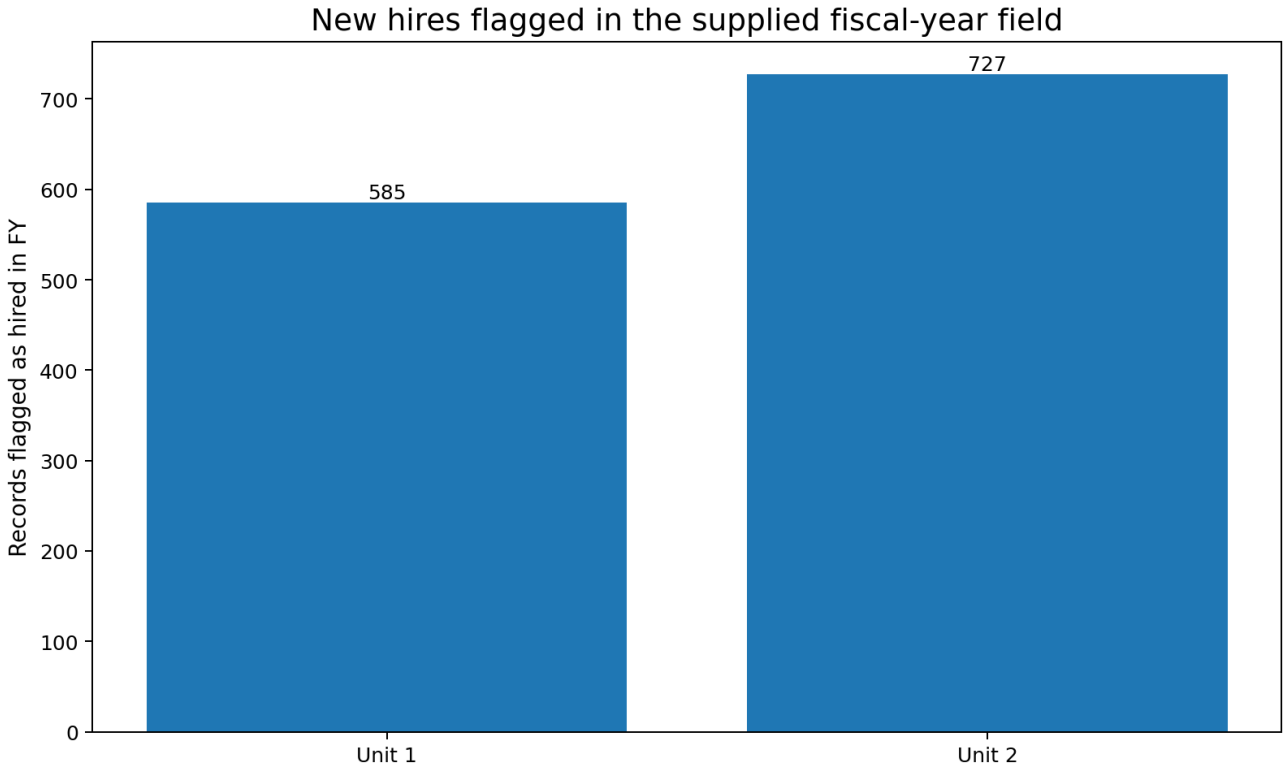
This chart stacks active and terminated records to compare workforce status counts. The figures are historical record counts, not an annual turnover rate.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

13. Bar chart — hires in FY

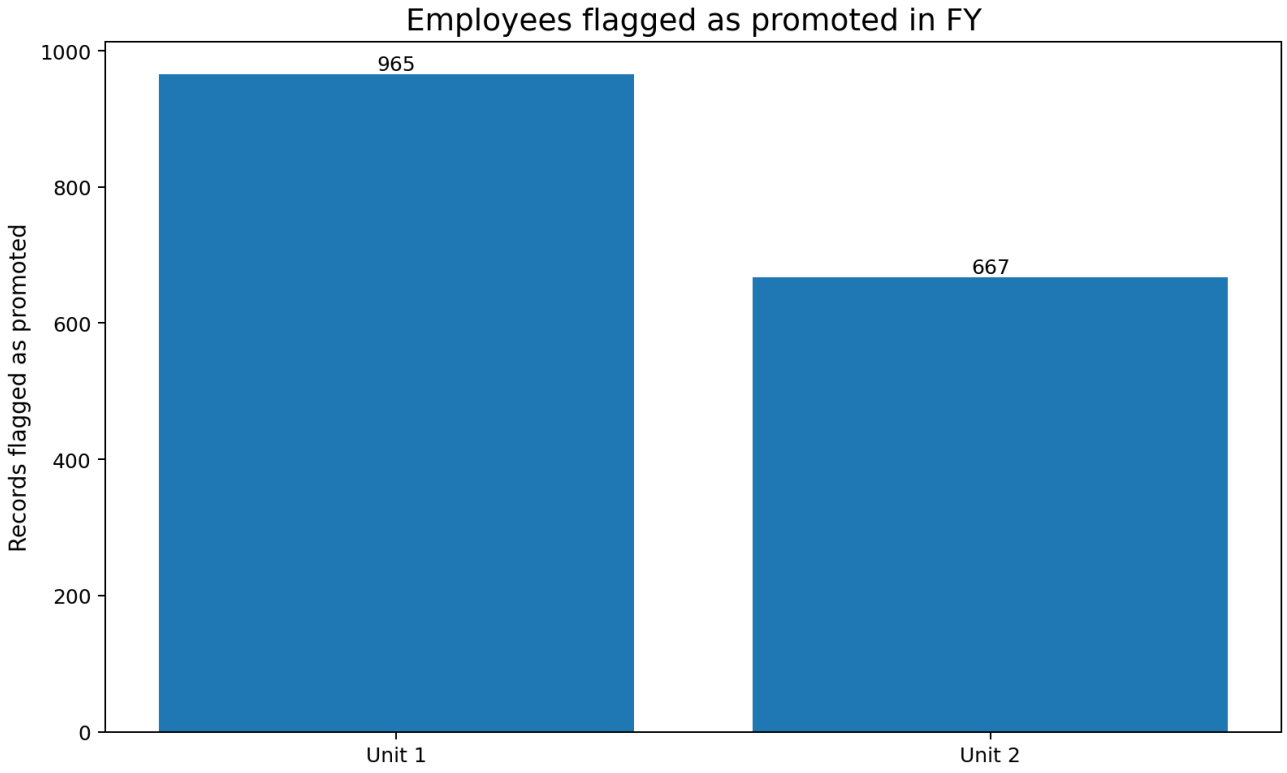
This chart counts records marked 'Yes' in the workbook's Hired in FY field. Validate the fiscal-year definition and date coverage before using it as a hiring rate.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

14. Bar chart — promotions

The chart counts records marked 'Yes' in Promoted in FY. It describes the supplied flag and should not be interpreted as a promotion rate without a validated eligible-population denominator.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

15. Boxplot — age by unit

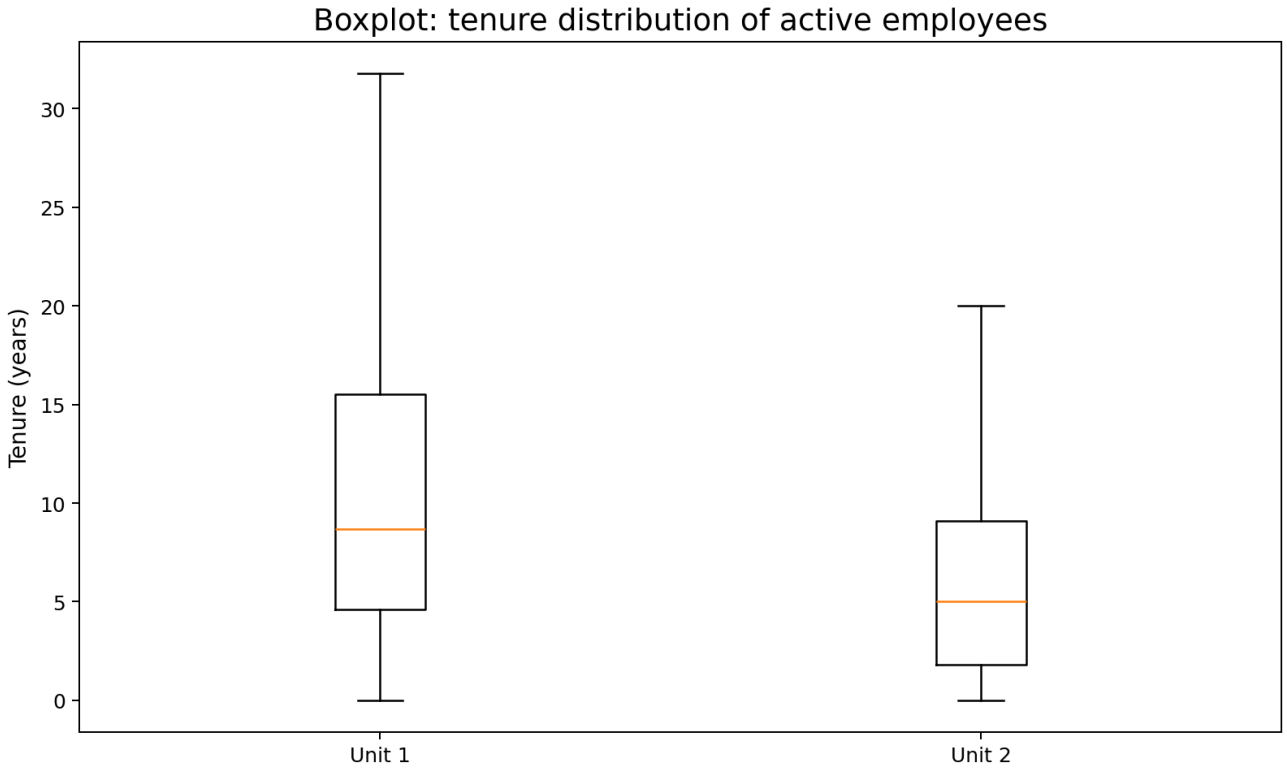
The boxplot compares medians and spread of active employee ages. Median age is 31.3 years in Unit 1 and 28.0 in Unit 2. Outliers are hidden to make the central distributions easier to compare.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

16. Boxplot — tenure by unit

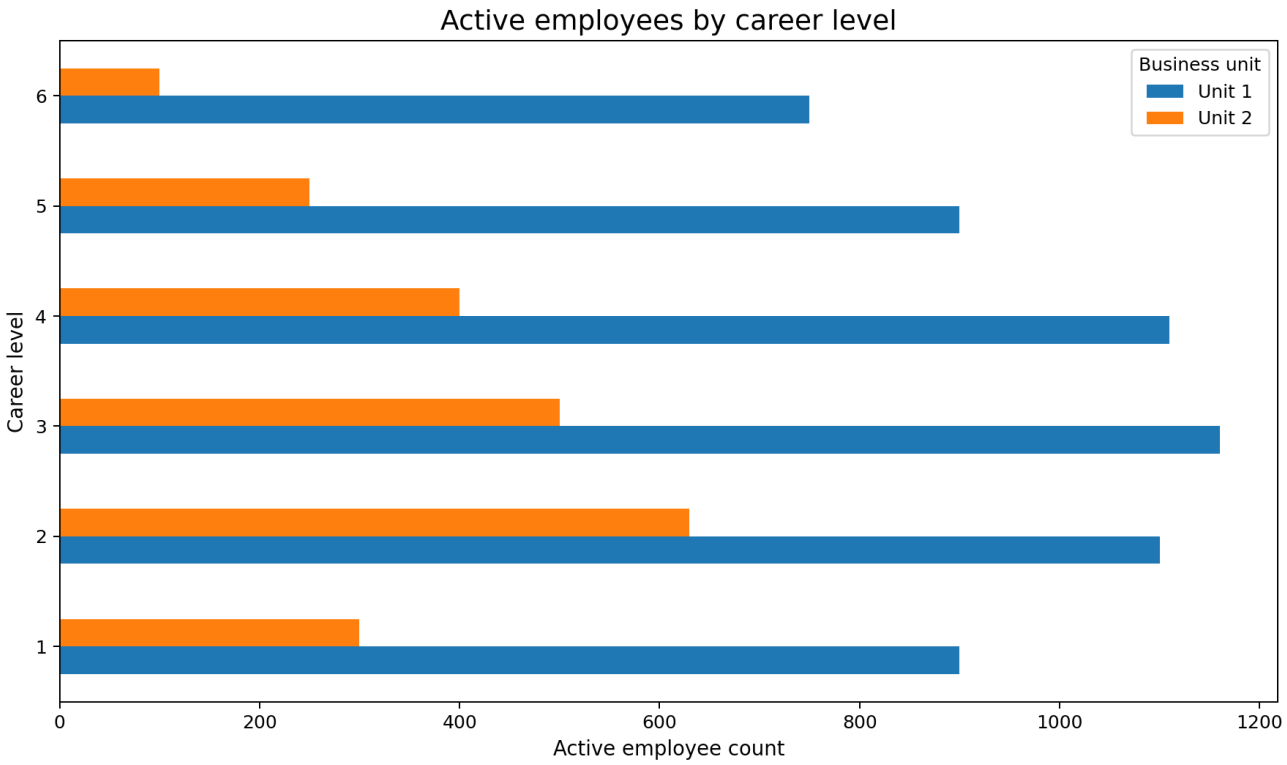
The boxplot compares the median and spread of service duration. Median tenure is 8.7 years in Unit 1 and 5.0 in Unit 2. This may inform experience and retention discussions, but not productivity conclusions.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

17. Horizontal bar — career-level mix

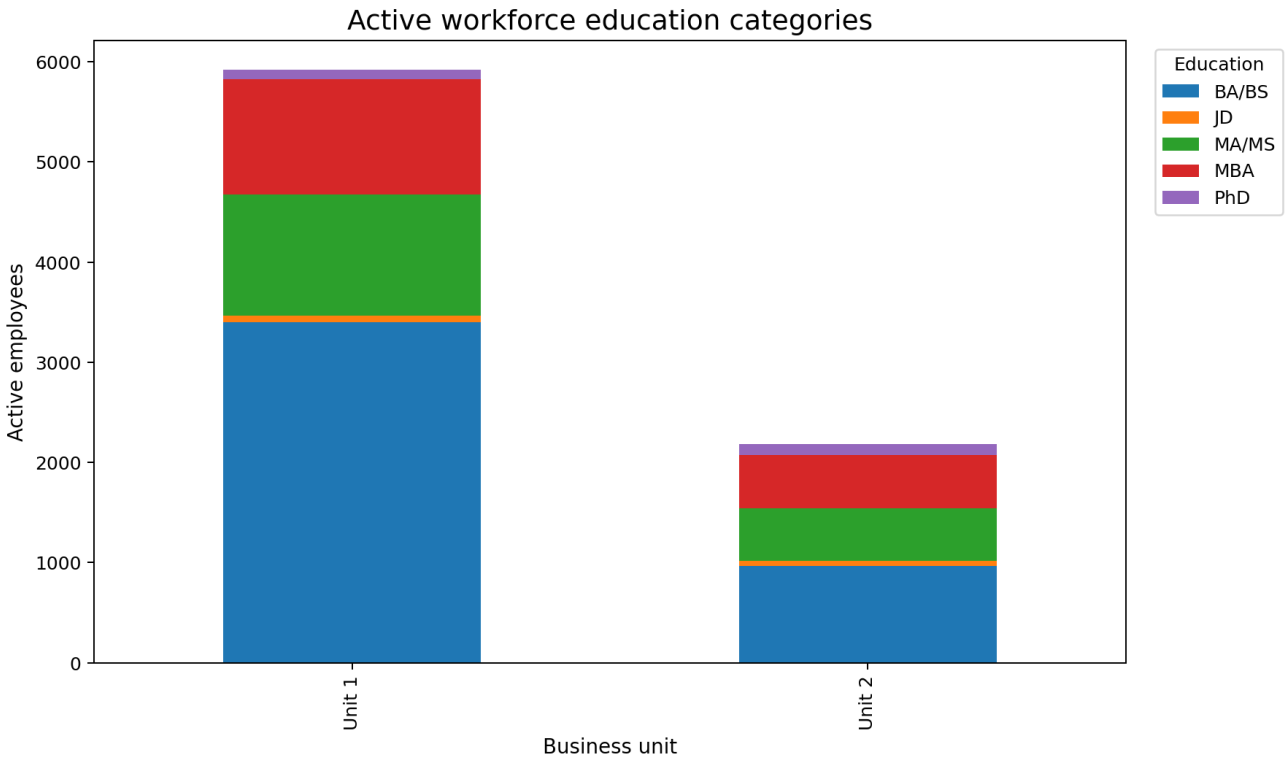
This view makes career-level counts easier to compare. Counts should be considered alongside workforce size; a large count does not imply role duplication or excess staffing.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

18. Stacked bar — education mix

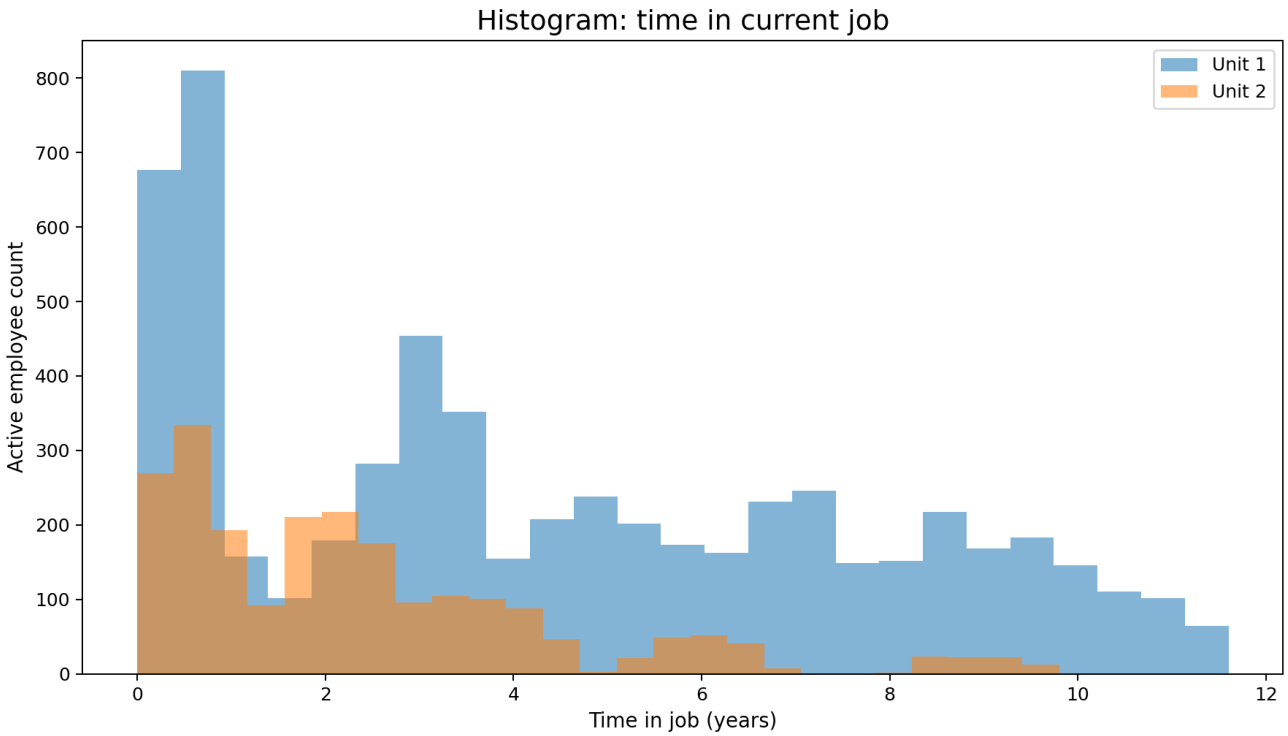
The chart compares recorded education categories for active employees. The workbook's category labels should be validated for consistent coding across units before drawing conclusions about qualification mix.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

19. Histogram — time in job

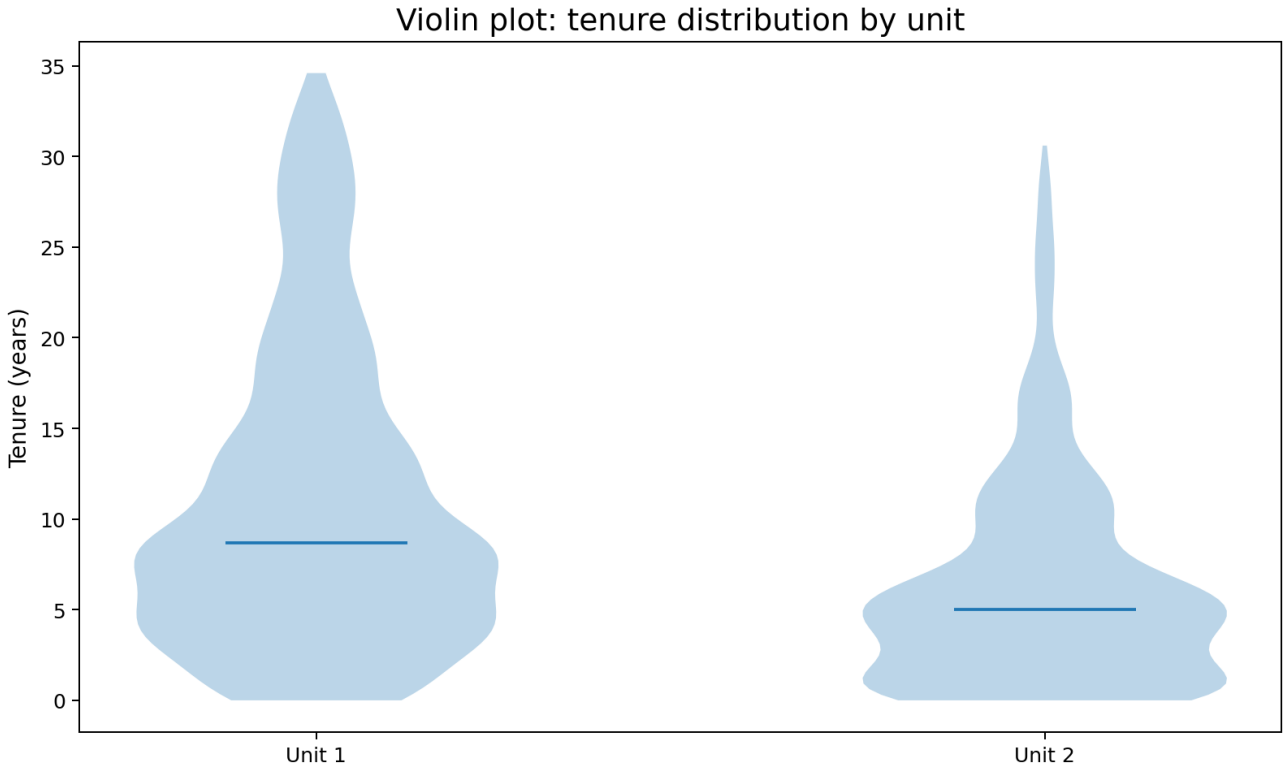
This histogram compares the time employees have been in their current job using Job Start Date. Longer time in job may reflect stability or limited mobility; interpretation requires role and career context.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

20. Violin plot — tenure distribution

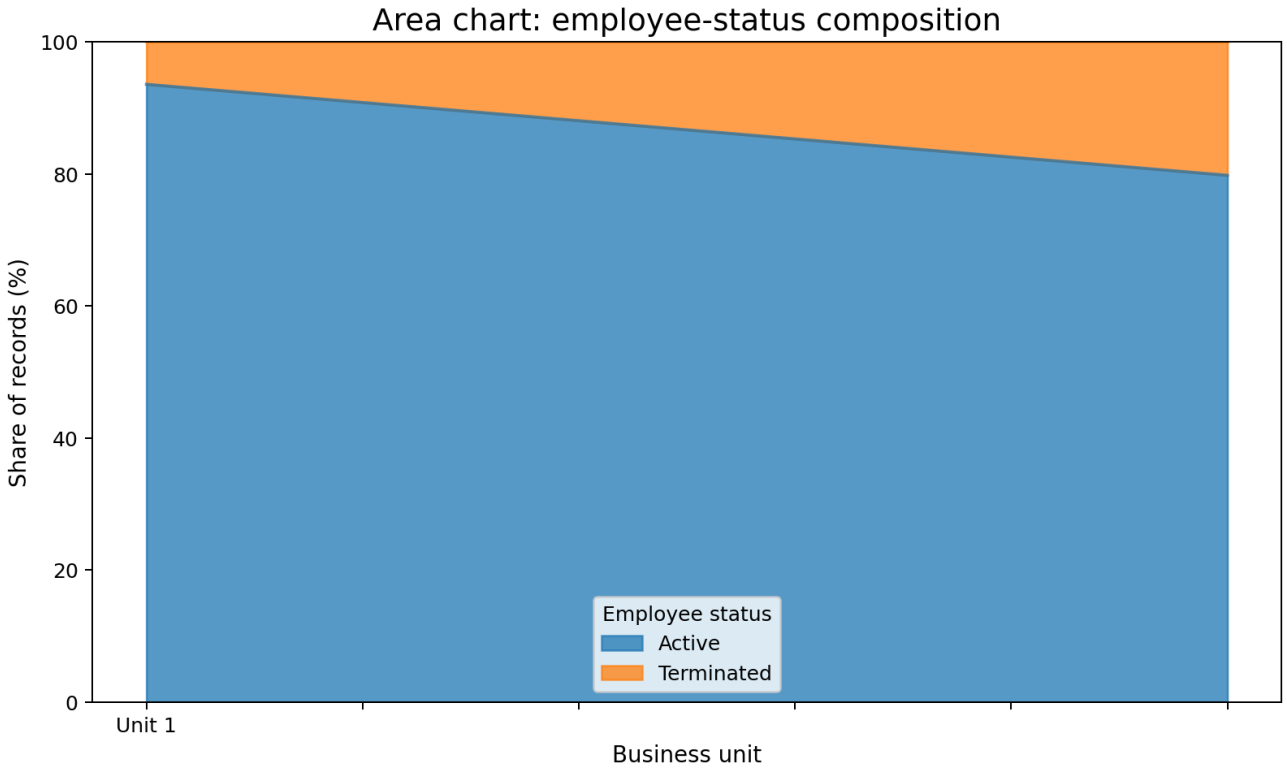
The violin plot shows the shape and density of tenure distributions, with median markers. It provides a different view from the histogram and boxplot; it does not explain why tenure differs.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

21. Area chart — status composition

The normalized area chart compares proportions rather than raw counts. Because it has only two business units, use it as a compact composition comparison, not as a trend over time.



Source: HR case-study workbook. Calculations use the described cleaning assumptions and 31 December 2016 reference date.

22. Conclusion and recommended decision approach

The charts provide complementary descriptive views of workforce size, status, age, tenure, career level, hiring, promotions, education and employee-record patterns. They identify areas for further investigation, but they do not establish why differences exist or whether roles are duplicative.

Recommendation: proceed with structured due diligence rather than immediate workforce reductions or final organization design. Validate the data, compare actual roles and responsibilities, obtain reliable FTE and compensation data, model scenarios, and protect critical talent before deciding on integration options.